

AACPDM 2026 Pre-Conference Course

Translational Research in Cerebral Palsy: How Emerging Science Will Transform Clinical Care

Philadelphia, PA | Fall 2026 | 4-Hour Course

PC1 - 9/27 8:00 AM – 12:00 PM ET

Course Description

Translational research in cerebral palsy and early onset developmental disorders is rapidly advancing. Discoveries in epigenetics, RNA biology, muscle physiology, and neural circuit function are poised to transform how we diagnose, treat, and counsel patients and their families. This course is designed to help practicing clinicians and members of the AACPDM community understand these emerging developments and their implications for patient care. Through a series of lectures, case discussions, and interactive workshops, participants will gain practical knowledge about how current translational research efforts may soon change clinical practice, and how they can engage with the emerging translational research programs.

Target Audience

This course is designed for clinicians and family members caring for children and adults with cerebral palsy and developmental motor disorders who want to understand emerging translational research and its clinical implications. The course will be valuable for pediatric neurologists, developmental pediatricians, physiatrists, orthopedic surgeons, neurosurgeons, neonatologists, and rehabilitation professionals. No prior research experience is required. Participants will leave with a clear understanding of how basic science discoveries are being translated toward clinical application and how to engage with research opportunities.

Learning Objectives

Upon completion of this course, participants will be able to:

1. Explain how perinatal brain injury leads to secondary changes in muscle and motor circuits that contribute to the evolving clinical picture in CP
2. Describe emerging biomarkers (epigenetic, RNA-based) that may enable earlier diagnosis and risk stratification in CP
3. Discuss with families and colleagues how current research on muscle stem cells and contracture biology may lead to new treatments
4. Apply standardized assessment approaches for dystonia that translate directly from research to clinical practice
5. Identify opportunities to participate in or refer participants to translational research studies
6. Discuss emerging neuroprotective and neurorestorative therapies with appropriate scientific context

Course Schedule

Format Key: [L] Lecture [P] Panel Discussion [C] Case Study [W] Interactive Workshop

Welcome and Expectations

Course Director Robert Akins, PhD
Nemours Children's Health

As attendees arrive, Dr. Akins will review the sessions and expectations. Explanatory slides will be displayed; explanatory handouts will be provided.

Session 1: The Brain-Body Connection in CP (60 minutes)

How does early brain injury lead to the motor problems we see in clinic?

Time	Speaker	Topic & Clinical Relevance
8:00-8:15	Lauren Jantzie, PhD <i>Kennedy Krieger/JHU</i>	[L] Encephalopathy of Prematurity: What Clinicians Need to Know How preterm birth and inflammation affect brain development; why cognitive and motor outcomes are linked; emerging therapeutic windows
8:15-8:30	Elizabeth Wright-Jin, MD, PhD <i>Nemours Children's Health</i>	[L] Immune Mechanisms in Neonatal Brain Injury Neuroinflammation as a target; maternal contributions; implications for risk counseling and advanced treatment
8:30-8:45	Shenandoah Robinson, MD <i>Kennedy Krieger/JHU</i>	[L] From NICU to Clinic: Translating Neuroprotection Research Current clinical trials in hydrocephalus prevention; neurorestorative strategies; what to tell families about available treatments
8:45-9:00	All Session 1 Faculty	[C] Case Discussion: The Preterm Infant at High Risk Interactive case: counseling families about brain injury, prognosis, and research participation; audience polling and discussion

Session 2: The Promise of Biomarkers (60 minutes): Can we diagnose CP earlier? Can we predict who will develop contractures?

Time	Speaker	Topic & Clinical Relevance
9:00-9:15	Robert Akins, PhD <i>Nemours Children's Health</i>	[L] Epigenetic Biomarkers: Toward Earlier CP Diagnosis What is epigenetic aging?; How muscle tissue tells the story of brain injury; DNA methylation as a potential diagnostic tool
9:15-9:30	Adam Marsh, PhD <i>University of Delaware</i>	[L] Machine Learning for CP Diagnosis: Promise and Pitfalls How computational approaches analyze complex biological data; what clinicians need to know about AI-driven diagnostics; validation requirements
9:30-9:45	Mona Batish, PhD <i>University of Delaware</i>	[L] RNA Biology and Muscle Development in CP Circular RNA as regulators of muscle formation; why muscles fail to grow normally in CP; potential for RNA-based therapeutics and diagnostics
9:45-10:00	All Session 2 Faculty	[P] Panel: When Will These Biomarkers Reach My Clinic? Timeline expectations; regulatory pathway; how clinicians can participate in validation studies; Q&A

***** BREAK – 15 Minutes *****

Session 3: Muscle, Spinal Cord, and Movement (60 minutes): *Why do muscles become stiff? Why is dystonia so hard to treat? What's coming?*

Time	Speaker	Topic & Clinical Relevance
10:15-10:30	Sudarshan Dayanidhi, PT, PhD <i>Shirley Ryan AbilityLab, Chicago</i>	[L] Why Muscles Fail in CP: Stem Cells, Sarcomeres, and Stiffness The biology of contractures; satellite cell depletion; mitochondrial dysfunction; implications for timing of surgery and rehabilitation; emerging regenerative approaches
10:30-10:45	Katharina Quinlan, PhD <i>University of Rhode Island</i>	[L] Spinal Cord Changes in CP: The Missing Link? How brain injury affects spinal motoneurons; parallels with other motor disorders (ALS, SMA); potential therapeutic targets at the spinal level
10:45-11:00	Bhooma Aravamuthan, MD, DPhil <i>Washington University</i>	[L] Dystonia in CP: From Neural Circuits to Treatment Why dystonia develops after brain injury; striatal cholinergic interneurons; standardized assessment methods you can use now; early treatment opportunities
11:00-11:15	All Session 3 Faculty	[W] Workshop: Assessing Movement Variability for Dystonia Hands-on: using video analysis to quantify leg movement variability; practice with sample videos; discussion of clinical implementation

Session 4: Bringing It Together (60 minutes): *How do I apply this knowledge? How can I get involved?*

Elapsed Contact Time	Speaker	Topic & Clinical Relevance
11:15-11:30	All Faculty	[C] Case Workshop: Integrating Translational Knowledge Small group case discussions: infant at risk for CP, child with progressive contractures, adolescent with dystonia; faculty split between groups; report back
11:30-11:50	All Faculty	[P] Panel: How to Engage with Translational Research Multi-site research networks; enrolling patients in studies; becoming a clinical collaborator; resources for clinician-researchers; funding opportunities
11:50-noon	Robert Akins, PhD <i>Course Director</i>	[W] Closing: Your Take-Home Action Plan Interactive: each participant identifies at least 2 specific ways to apply course content; resource handout distribution; course evaluation

Faculty

Course Director

Robert Akins, PhD — Principal Scientist, Nemours Children's Health System. Over 30 years of research experience specializing in epigenetics, neuromuscular junction biology, and cerebral palsy biomarker development.

Faculty (alphabetical)

Bhooma Aravamuthan, MD, DPhil — Assistant Professor of Neurology, Washington University School of Medicine; Director, Cerebral Palsy and Mobility Center, St. Louis Children's Hospital. Nationally recognized expert in dystonia in CP; recipient of the AAN Jon Stolk Award and CNS Philip R. Dodge Young Investigator Award.

Mona Batish, PhD — Associate Professor, Medical and Molecular Sciences, University of Delaware. NIH Director's Early Independence Award recipient; expert in single-molecule RNA imaging and circular RNA biology.

Sudarshan Dayanidhi, PT, PhD — Research Scientist, Shirley Ryan AbilityLab, Chicago. Leading researcher on muscle stem cells in CP; author of key papers on contracture biology and mitochondrial function in spastic muscle.

Lauren Jantzie, PhD — Associate Professor of Pediatrics, Neurology, and Neurosurgery, Johns Hopkins University School of Medicine; Kennedy Krieger Institute. Expert in encephalopathy of prematurity; leading clinical trials in hydrocephalus prevention.

Adam Marsh, PhD — Associate Professor of Marine Science and Bioinformatics, University of Delaware; Co-Founder, Genome Profiling LLC. Expert in computational analysis of DNA methylation and machine learning approaches to disease diagnosis.

Katharina Quinlan, PhD — Associate Professor, Biomedical and Pharmaceutical Sciences, University of Rhode Island; Ryan Institute for Neuroscience. Expert in spinal motoneuron physiology in motor disorders including CP, ALS, and SMA.

Shenandoah Robinson, MD — Professor of Neurosurgery, Johns Hopkins Children's Center; Kennedy Krieger Institute; President, American Academy of Neurological Surgery. Expert in surgical treatment of spasticity and epilepsy; leading clinical trials in hydrocephalus treatment.

Elizabeth Wright-Jin, MD, PhD — Pediatric Neurologist, Nemours Children's Health; Assistant Professor, University of Delaware. Expert in neonatal hypoxic-ischemic encephalopathy and immune-brain interactions; developer of novel mouse models of HIE.

Teaching Methods

This course uses multiple teaching formats to maximize engagement and clinical applicability:

- Focused lectures (15-20 minutes) emphasizing clinical relevance
- Interactive case discussions with audience participation
- Panel discussions with Q&A
- Hands-on workshop (dystonia assessment)
- Small group exercises with faculty facilitation

Course PDFs

Scan the QR Code or click on it to go to the link (available until 12 October 2026)

